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First Palaeogene sedimentary rock palaeomagnetic pole from stable western Eurasia and tectonic implications

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SUMMARY

A palaeomagnetic investigation of lower Eocene (ca. 52 Ma) London Clay Formation cemented mudstones from Sheppey (SE England) has yielded a mean direction of Dec. = 1.1°, Inc. = 43.2°, where $N = 9$, $\alpha 95 = 6.8°$ and $K = 58.5$. This apparently high-quality direction (Q-factor = 5) has an associated palaeopole of 178.6°E, 63.7°N, where $A_{95} = 6.8°$. The data represent the first pole from post Jurassic stable Eurasia rocks outside of the European North Atlantic Igneous Province (NAIP), of which most results have been obtained from NW Britain and the Faroe Islands. The data can in part be used to constrain the position of Palaeogene Eurasia, in particular the zero-offset declination implying negligible rotation of western Eurasia since the early Cenozoic. This is in contrast with data derived from the European NAIP, which imply small to moderate clockwise rotations for this part of the plate. The inclination angle may provide less useful information as it appears to be anomalously shallow when compared with that associated with the NAIP derived poles. In an attempt to understand the shallowing, we re-examined data from Palaeocene–Eocene sediments recovered in several boreholes (bathyal sediments in DSDP Hole 550, four cores through fluvio-delatic to middle shelf sequences in the London area, and one borehole sequence from East Anglia). In all cases, the sediments show systematic inclination shallowing similar in magnitude to that reported from Sheppey. Tectonic and geomagnetic explanations can be discounted; sediment compaction appears to be the likely cause. In light of the current controversy surrounding the 'stable Asia shallow inclination problem', the result reinforces the suggestion that tectonic modelling needs to be done carefully when the supporting data are based exclusively on palaeomagnetic studies of sedimentary rocks.

Key words: Eocene, Eurasia, London Clay Formation, North Atlantic Igneous Province, palaeomagnetism, shallow inclinations.

INTRODUCTION

Incredibly, perhaps, the position of 'stable' Late Cretaceous–Palaeogene Eurasia is based largely on palaeomagnetic data from the British sector of the North Atlantic Igneous Province (NAIP), at the westernmost tip of Earth's largest continental plate (Fig. 1). The importance of the NAIP poles is increased because outside of this time window there are no Cretaceous and Cenozoic rock-derived palaeomagnetic poles from the stable part of the plate. Most of the preserved lithologies are unconsolidated sediments and because they cannot be core drilled they are considered inappropriate for tectonic studies (Torsvik *et al.* 2001).

Recently, Riisager *et al.* (2002) published the first new palaeopole from the NAIP, 16 years after Dagley & Mussett (1986) obtained the last one from Muck and Eigg in western Scotland. A key feature of the new Faroe Islands pole (71.4°N, 154.7°E, where $A_{95} = 6.0°$), from basaltic flows erupted during Chrons C26n (~57.7 Ma) to C24r (55 Ma), is its offset with respect to the Torsvik *et al.* (2001) 60 and 55 Ma averaged NAIP poles (77.4°N, 165.8°E, where $A_{95} = 2.8°$). This prompted Riisager *et al.* to suggest that there could be problems with the existing NAIP-pole database. First, all but two of the 'reliable' poles used in the Torsvik *et al.* (2001) compilation were generated before 1980, and in many of the studies the methods used in processing the material (e.g. one-step blanket demagnetization) and calculation of characteristic remanence directions (ChRM) without principal component analyses (Kirschvink 1980) would now be considered insufficient/inappropriate. Secondly, several of the NAIP poles have 95 per cent confidence circles that do not overlap when, in fact, they should do. Thirdly, many of the NAIP poles are linked to the geological time-scale using radiometric dates

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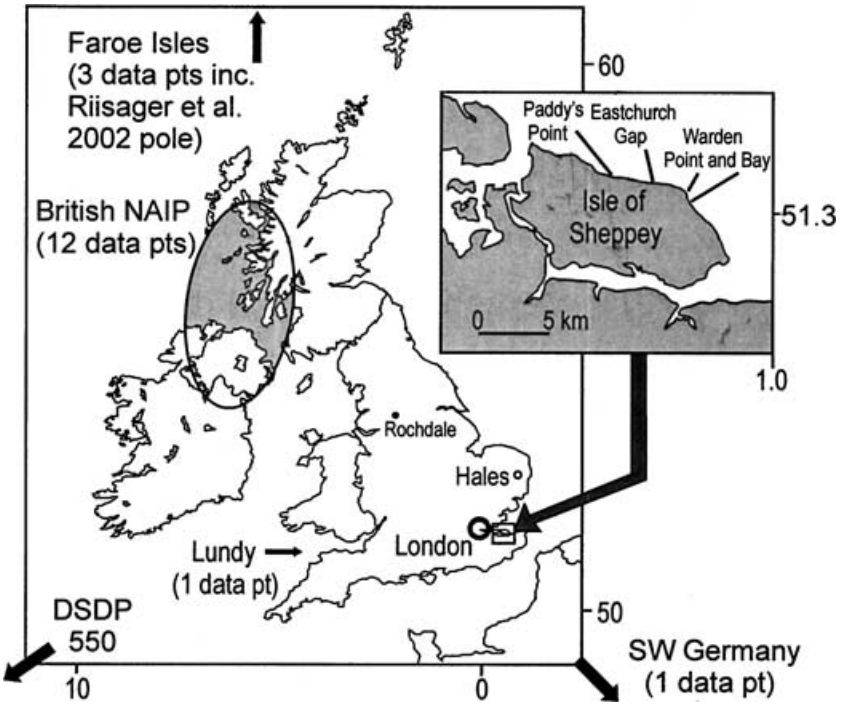


Figure 1. Map of the sections sampled on Sheppey and the approximate area from which the twelve 'reliable' (Torsvik *et al.* 2001) British NAIP apparent poles have been obtained. The other 'stable Europe' data are from the Faroe Isles; Lundy Island, Bristol Channel, UK; and southwest Germany. The locations of other sections discussed in the text are also shown.

cited in the original works, but in many cases newer more reliable Ar–Ar and/or biostratigraphic dates exist (e.g. compilation of Saunders *et al.* 1998, their fig. 7) but these have not been incorporated into the 2001 IAGA Palaeomagnetic Database.

In an attempt to throw light on the issue, we have carried out an investigation of sedimentary rocks from outside of the European NAIP, specifically 10 sites in cemented lower Eocene (ca. 52 Ma) middle shelf sediments from the Isle of Sheppey, southeast England (Fig. 1). Theoretically, the rocks should provide a good western Eurasia early Eocene apparent pole. First, they were deposited at a site almost 1500 km from the NE Atlantic Ypresian spreading centre (Larsen & Saunders 1998, their fig. 1) and should not be disturbed by this tectonic activity. Secondly, they are situated beyond the northern limit of the Alpine deformation front. Thirdly, they have not suffered cryoturbation, as is the case for some similar-age sequences in Scandinavia (Heilmann-Clausen *et al.* 1985). Fourthly, their precise middle-late early Eocene age (Chron C24n.1n-C23r-C23n.2) is known following detailed magnetobiostratigraphic studies (Ali *et al.* 1993).

GEOLOGY AND PREVIOUS PALAEOMAGENTIC WORK

The Isle of Sheppey is located about 70 km east of central London (Fig. 1). Because of the lower Eocene London Clay Formation's rich fossil content, the 6-km-long series of cliff and beach sections along Sheppey's northern coast have long attracted the attention of geologists (Prestwich 1847, 1854, 1888; Whitaker 1872). However, despite more than 150 yr of investigation, the detailed stratigraphy of the London Clay on Sheppey remained incompletely understood until King (1984), who described a 53.55-m-thick sequence of intensively bioturbated silty-clays and clayey-silts that belonged to the upper third of the formation. In developing his scheme, King made use of a series of persistent 20 to 30-cm-thick calcareous nodule bands, labelled A–P, to subdivide what is an otherwise monotonous sequence of unlithified sediments.

During the original palaeomagnetic studies on Sheppey (Ali 1989), the nodule bands were never drill core sampled and instead the unconsolidated sediments were collected using the 'copper tube' method developed by Townsend & Hailwood (1985). Although the technique is perfectly adequate for polarity studies, the data were never used to calculate apparent poles. The consensus amongst E.A. Hailwood's 1980s research group was that the samples possibly carried a slightly disturbed remanence, acquired during the tube insertion/core extrusion, and that it was better to only use the data for magnetostratigraphic purposes. Using dinoflagellate data (Islam 1983), the N–R–N polarity sequence identified at Sheppey was correlated with the now-labelled Chrons C24.1n, C23r and C23n.2n (Ali *et al.* 1993). This correlation was further strengthened following a related study of an almost identical succession in western Belgium-northern France (Ali *et al.* 1993), where a detailed calcareous nannofossil biostratigraphy was also available (Steurbaut & Nolf 1986).

SAMPLING AND METHODOLOGY

Using a gasoline-powered rock drill, six or seven 2.5-cm diameter cores were sampled from nine of the main septarian nodule horizons (A–G, J and P; horizons H, K, L, M and N were either too friable or simply inaccessible). The cores were oriented using a Brunton Compass mounted on a Pomeroy Orientation Table. To supplement the data set, an additional site was collected from a less persistent nodule band approximately 2 m above nodule band G. Bands A–E are exposed at Warden Point (Fig. 1), where the section dips towards the NW at about 0.5°; F, G and G + 2 m are exposed

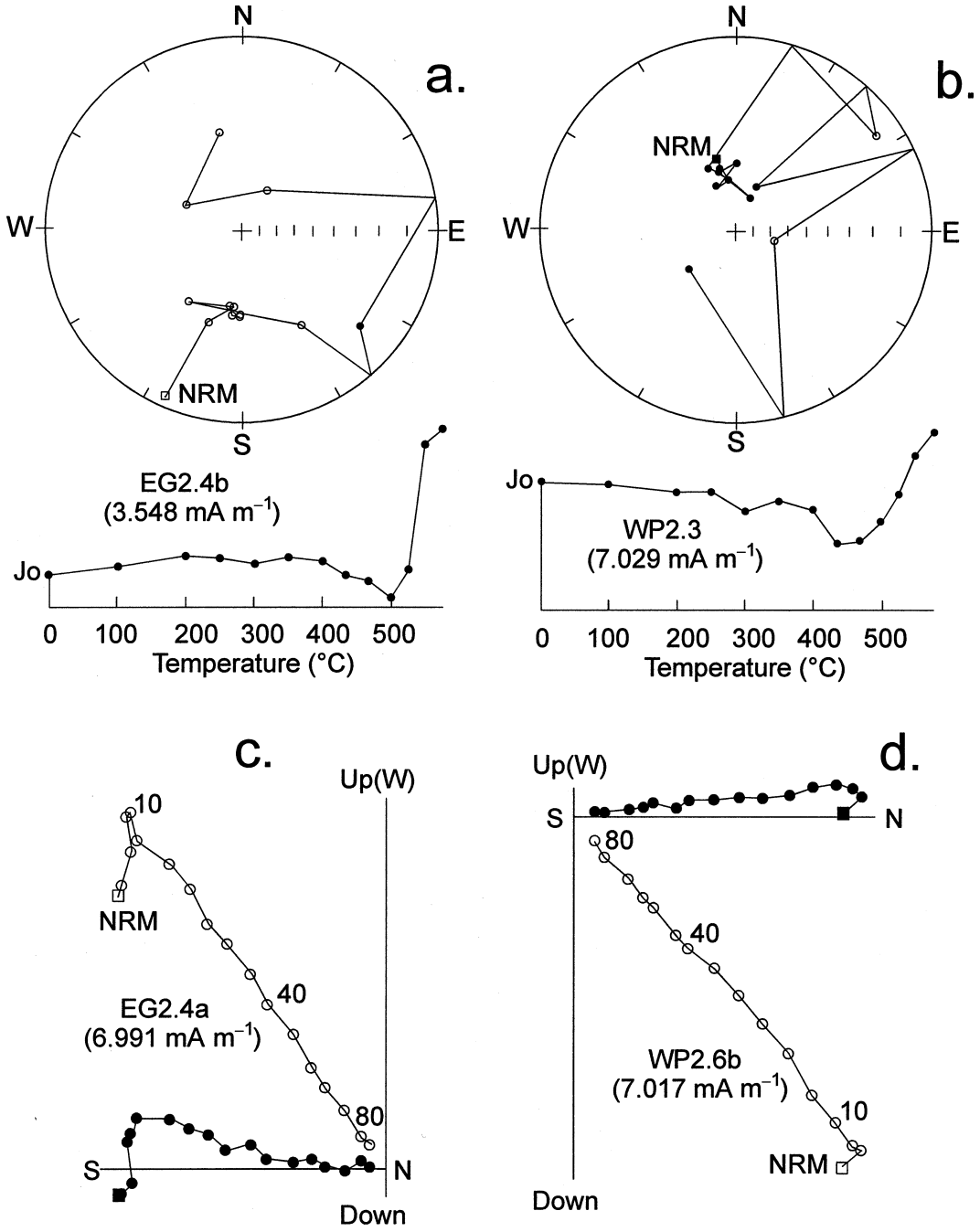


Figure 2. Examples of demagnetization. (a) and (b) are stereonet/intensity decay plots for thermally demagnetized samples. The technique is not suitable for the Sheppey rocks as the samples show large intensity increases and directional changes above 400 °C. (c) and (d) are vector-endpoint plots showing the much better response of samples (from the same sites) to af processing. The numbers indicate the applied field in mT; solid/open symbols are plotted on the lower/upper hemispheres of the stereonet plots and horizontal/vertical projection planes of the vector-endpoint diagrams.

at Eastchurch Gap, where the bedding is horizontal; J and P are exposed at Paddy's Point, where the succession dips towards the NW at about 0.5°.

The natural remanent magnetization (NRM) of each sample was measured using an AGICO JR5A magnetometer. The specimens were magnetically cleaned using alternating field (af) demagnetization in a Molspin tumbler system. [Six pilot samples were processed using thermal demagnetization, however, all showed large-scale intensity increases and directional jumps above c. 433 °C (Figs 2a and b), and it was impossible to define characteristic magnetizations, ChRM. The likely cause of this behaviour is heating-induced sulphide (pyrite) reactions – the London Clay Formation at Sheppey has an appreciable pyrite content, mainly in the form of infilled burrows, see King (1984, p. 147).] Vector-endpoint diagrams (Zijderveld 1967) were used to analyse the demagnetization data and principal component analysis (Kirschvink 1980) was used to calculate ChRM directions. All 10 sample sites carry a high-stability remanence with either no other component or an easily removed

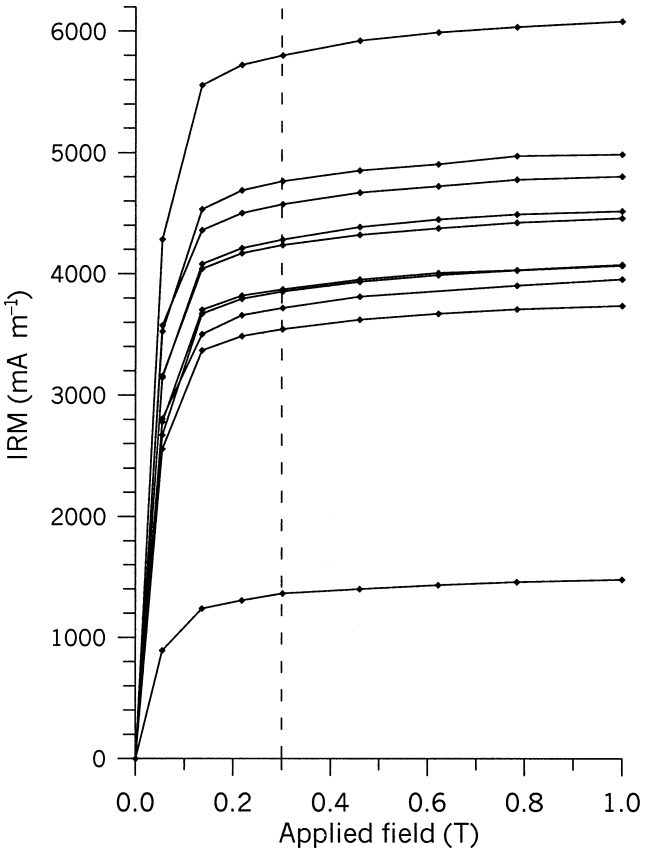


Figure 3. IRM acquisition curves for representative specimens from each of the sampled nodules. The dashed line at 0.3 T is useful for estimating IRM ratios (Ali 1989).

viscous remanence (e.g. Figs 2c and d), indicating a simple magnetization history.

Unfortunately the thermal demagnetization experiments provided little useful information on the remanence carriers. Such information was garnered using isothermal remanent magnetization (IRM) acquisition experiments. All of the processed specimens (one per site) reached above 92 per cent of the saturation IRM by 0.3 T (Fig. 3, Table 1), suggesting that the remanence is carried by a low-coercivity magnetic mineral. The relatively high peak IRM values hint at this mineral being magnetite. Fisher (1953) statistics were used to calculate site mean directions (Table 1). Respectively, the results are presented as a composite stratigraphic log (Fig. 4) and on stereonet plots (Figs 5 and 6).

RESULTS

The obvious feature to emerge from Fig. 4 is that the core-drilled succession records a N–R–N polarity sequence. In detail there is a slight difference with the findings of Ali *et al.* (1993), the Chron C24n.1n/C23r reversal being placed at 2.0 m in the section (between nodule bands B and C), which contrasts with the earlier study where the boundary was placed at 3.55 m. The lack of drill-sample sites close to the Chron C23r/C23n.2n transition means that the boundary should remain positioned at 25.1 m. A noteworthy point is that Site PP2 (at 18.45 m), from the lower part of the section at Paddy Point, records an anomalous direction (Dec. = 159.7°, Inc. = 1.3°), and probably marks a geomagnetic excursion (grey shade in Fig. 4).

Table 1. Summary of palaeomagnetic data obtained from Sheppey. Stratigraphic heights based on King (1984). Nc (Np) = number of specimens from a site used to calculate (Nc) the mean direction from the number of specimens processed (Np). Intensity values = the lower/upper initial intensity values within a site. In the 'demag. column,' af/th = alternating field/thermal treatment, f = thermal treatment failed; Those sites used to calculate the outcrop mean are highlighted with the ⊗ symbol (?E = site recording a possible geomagnetic excursion).

Site	Nodule band	Height in section (m)	Nc(Np)	Intensity (mA m^{-1})	Demag (af, th[f])	*IN SITU* Dec.	*IN SITU* Inc.	TILT CORR. Dec.	TILT CORR. Inc.	POL	A-95 /AS(*)	K	IRM ratio	Peak IRM (mA m^{-1})
Locality = Paddy's Point; N 51°25′ 30.7″; E 0°49′ 59.7″; Strike = 225°; Dip = 0.5°; Dip direction = NW														
PP1	P	36.90	7 (7)	2.1–4.3	af, th(f)	19.1		18.5	55.0	N	3.8	259.5 ⊗	0.94	3956
PP2	J	18.45	7 (7)	1.7–4.9	af	159.7	0.8	159.7	1.3	R	12.9	23.0 ?E	0.95	4069
Locality = Eastchurch Gap; N 51°25′ 14.8″; E 0°52′ 05.8″; Structural correction = none														
EG3	G+2 m	14.30	1 (7)	1.6–4.3	af	186.0	−38.4	186.3	−38.1	R	NA	NA ⊗	0.92	1480
EG2	G	12.30	7 (7)	3.6–7.0	af, th(f)	183.5	−49.3	183.5	−49.3	R	6.0	101.8 ⊗	0.95	4076
EG1	F	8.60	7 (7)	4.1–7.3	af, th(f)	185.1	−29.6	185.1	−29.6	R	7.8	61.0 ⊗	0.95	6082
Locality = Warden Point; N 51°24′ 53.7″; E 0°54′ 13.2″; Strike = 225°; Dip = 0.5°; Dip direction = NW														
WP5	E	5.40	7 (7)	1.9–10.9	af	172.4	−45.1	172.1	−44.7	R	7.0	78.8 ⊗	0.95	4459
WP4	D	4.30	8 (8)	1.8–4.5	af, th(f)	174.0	−32.5	173.8	−32.1	R	13.7	17.3 ⊗	0.95	3737
WP3	C	2.80	7 (7)	1.4–4.0	af, th(f)	185.4	−39.2	185.1	−38.9	R	8.7	48.6 ⊗	0.95	4516
WP2	B	1.20	6 (6)	5.0–8.9	af, th(f)	357.3	50.6	356.9	50.2	N	4.5	218.6 ⊗	0.96	4987
WP1	A	0.00	5 (5)	4.1–13.9	af	351.2	48.4	350.9	48.0	N	8.7	78.8 ⊗	0.95	4803
Mean RP Sites	WP3-EG3		6			181.1	−39.2	181.0	−38.9	R	7.4	83.2		
Mean NP Sites	WP1-2 + PP1		3			1.7	52.0	1.3	51.6	N	14.7	71.5		
Mean 9 sites IS	Excl. PP2		9			1.3	43.4				6.8	58.6		
Mean 9 sites TC	Excl. PP2		9					1.1	43.2		6.8	58.5		

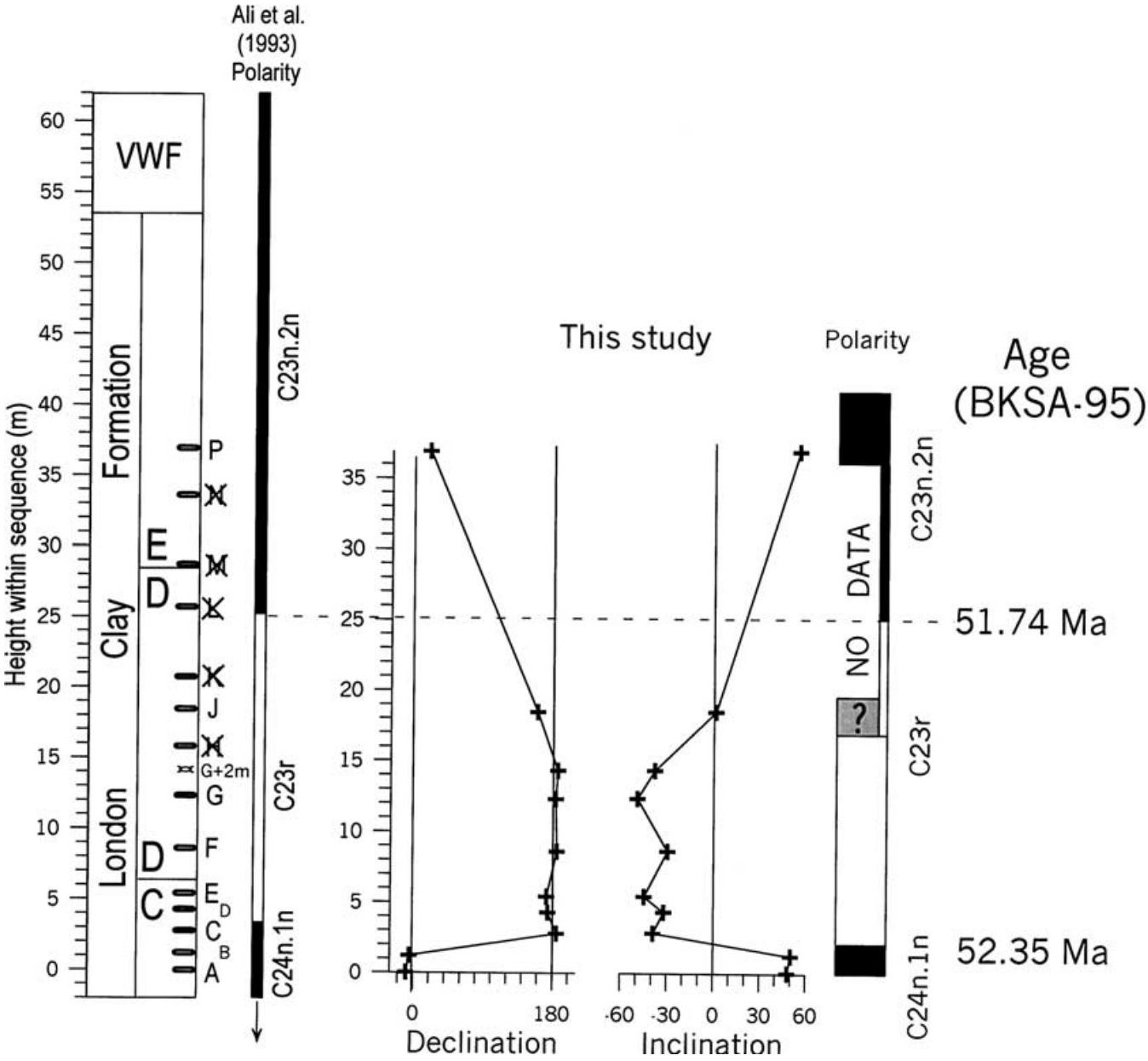


Figure 4. Remanence data from Sheppey in a stratigraphic context. C/D and D/E denote King's (1984) divisional boundaries; A–P are the nodule horizons, H and K-N could not be sampled; VWF is the Virginia Water Formation. BKSA-95 refers to the Berggren *et al.* (1995) time-scale.

In terms of tectonic studies, nine of the 10 sites provide apparently useful directional data (Fig. 5), although at Site EG3 only one from the seven processed samples yielded a ChRM. There appears to be a slight bias in the data with the mean inclination of the three normal polarity sites (51.6°) being somewhat steeper than the six reverse polarity sites (−38.9°) (Table 1). However, the angular separation of the two data subsets is 12.7° (Table 1) and the 95 per cent confidence circles show a considerable overlap (Fig. 6). Application of the McFadden & McElhinny (1990) reversal (simulation) test is positive (Rc, where observed $\gamma = 12.7°$, with a critical value of 15.55°). The normal polarity corrected mean direction of the nine sites is Dec. = 1.1°, Inc. = 43.2°; where $\alpha 95 = 6.8°$ and $k = 58.5$, and this yields an apparent pole with a longitude of 178.6°E and latitude of 63.7°N.

Van der Voo's (1990) scheme provides a rigorous method for assessing the quality of a palaeomagnetic direction/pole. The Sheppey data have a Q-factor of 5 (from a maximum of 6; they cannot be tested for the 'similarity to younger poles' category since no data exist for a comparison to be made). Thus the obtained direction appears to be reliable, *cf.* Torsvik *et al.* (2001) European compilation, where the mean value is 4.76 ± 1.03.

COMPARISON WITH THE NAIP POLES

Comparison of the Sheppey pole with those obtained from the NAIP (Fig. 7) raises a number of noteworthy points. First the pole plots almost exactly on the opposite longitude of the globe to the sampling location implying no rotation of the site and by inference no rotation of westernmost Eurasia since the early Eocene. The Torsvik *et al.* (2001) pole hints at a slight clockwise rotation, although the confidence circle, which is rather small, almost overlaps the 180° meridian. In contrast the supposedly superior Riisager *et al.* (2002)

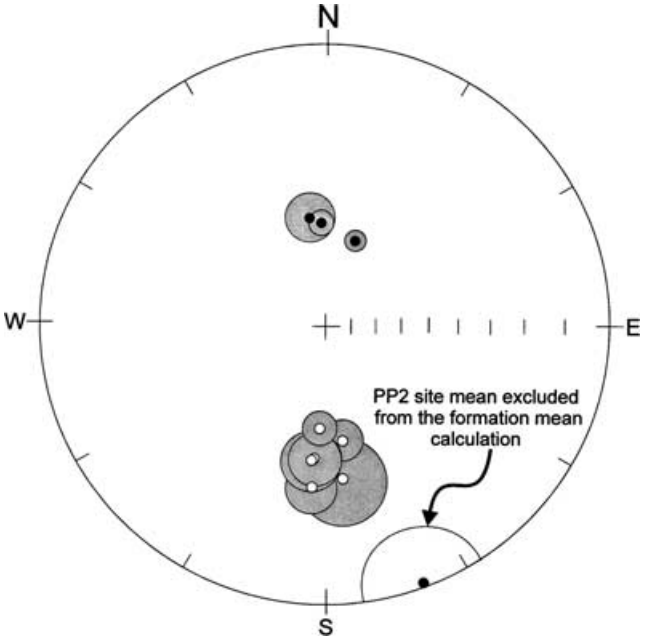


Figure 5. AF cleaned mean directions of sites with 95 per cent confidence circles from Sheppey plotted on an equal angle stereonet. Solid/open circles are downward/upward dipping.

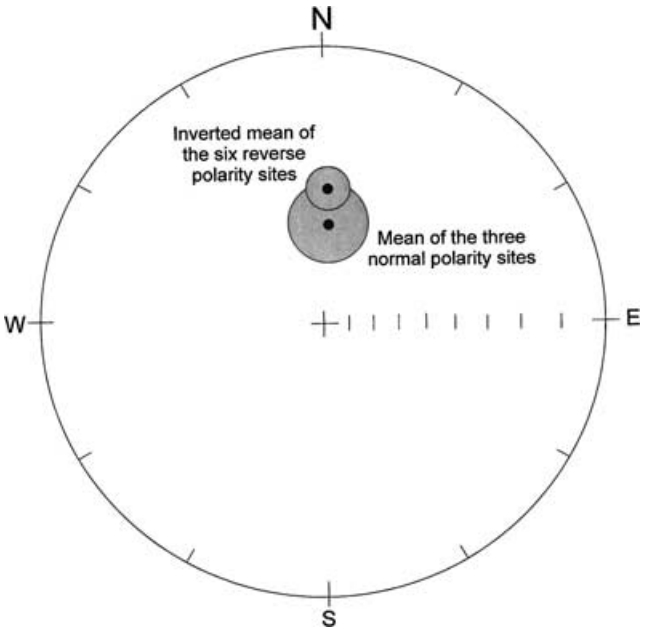


Figure 6. Means of the three normal (Sites WP1-2, PP1) and six inverted reversed polarity (Sites WP3-5, EG1-3) site data plotted on an equal angle stereonet. Site PP2 has been excluded.

Faroe Islands pole sits more than 25 degrees of longitude away from 180° and the inescapable conclusion is that the sample site and/or western Eurasia underwent a small but notable clockwise rotation. Such an observation is non-trivial because the NAIP poles are used to position southern and eastern Asia's Palaeogene active margins, which in turn impacts, for example, on the India–Eurasia collision (e.g. Klootwijk *et al.* 1994; Li *et al.* 2002). It also impinges on an ongoing debate concerned with the shallow inclinations recorded in Cretaceous–Cenozoic red-beds in 'stable' Asia (Cogne *et al.* 1999; Gilder *et al.* 2001; Si & Van der Voo 2001; Bazhenov & Mikolaichuk

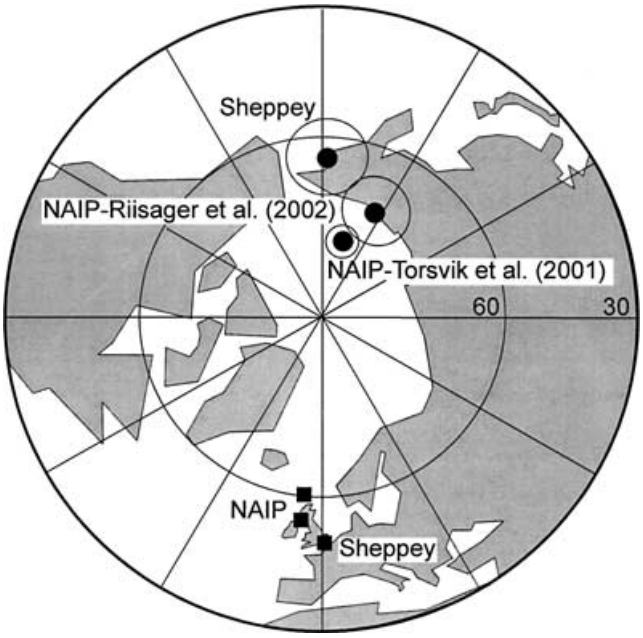


Figure 7. Apparent pole derived from Sheppey as compared to the European NAIP poles of Torsvik *et al.* (2001) and Riisager *et al.* (2002). Squares/circles indicate the sampling sites/apparent pole positions.

2002; Dupont-Nivet *et al.* 2002). The new Faroe pole implies an even greater degree of shallowing than is currently the case with the older averaged NAIP pole.

The second obvious feature of the Sheppey data is that the inclinations are anomalously shallow resulting in an apparent pole located at a latitude lower than those obtained from the NAIP (Fig. 7). In an attempt to understand this feature, previously published inclination data from three key upper Palaeocene–lower Eocene sites relatively close to the NAIP were examined, and were chosen because the cored sequences provide a range of depositional facies, from bathyal to fluvial-deltaic. The one drawback for such a study is that it is not possible to construct a palaeopole from these borehole sample data (because of the lack of declination control). However, plotting of the inclination data (Fig. 8) in which the predicted inclination values, based on Torsvik *et al.* (2001) and Riisager *et al.* (2002) NAIP poles are also shown, is most informative.

The upper Palaeocene–lower Eocene bathyal sequence recovered in DSDP Hole 550, drilled on the Goban Spur in 1980, has recently been the subject of magnetostratigraphic reinvestigation (Ali & Hailwood 1998). Data from that study can be recycled for this investigation. From Fig. 8(a), it is clear that the majority of samples have inclinations that are appreciably lower than the NAIP derived values. Note that Ali & Hailwood (1998) drew attention to the slightly shallower negative inclinations in samples from this hole when compared to those with positive values, but the overwhelming majority of positively inclined specimen directions also have angles well below the predicted values (55.4°/49.3°).

The Hales borehole, Norfolk, recovered a 55-m-thick Palaeogene middle shelf sequence that in part records Chrons C26r, C26n, C25r and C24r (no C25n) (Ali 1989; Ali & Jolley 1996). Fig. 8(b) shows the inclination data for samples with clearly defined stable-endpoint directions. As with DSDP Hole 550 and the four London area cores, the specimens have ChRM inclinations clearly lower than predicted absolute values (59.4° or 54.6°).

Ellison *et al.* (1996) presented results from a magnetostratigraphic investigation of four borehole cores from lower part of the London

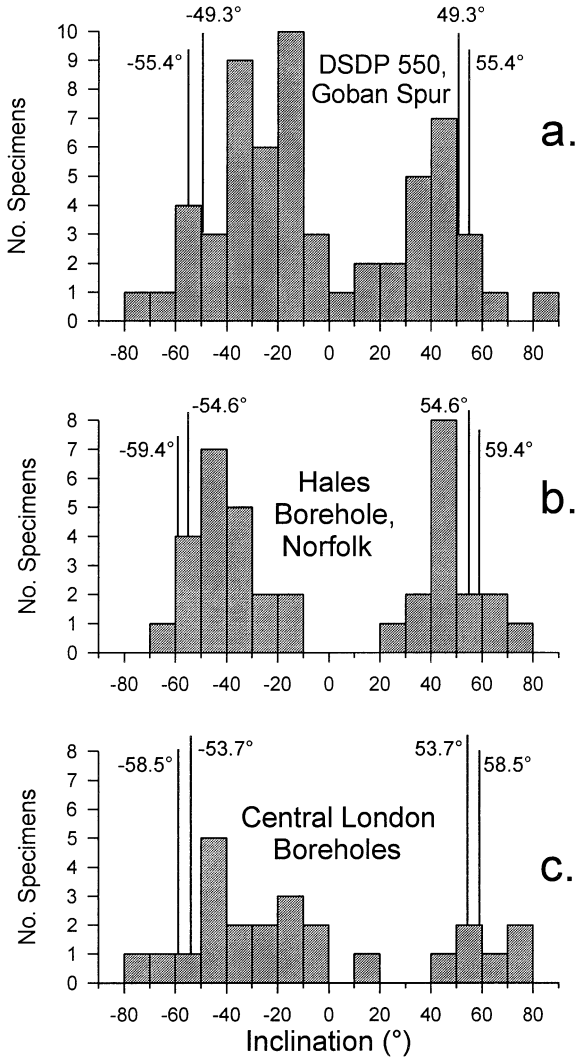


Figure 8. Inclination data from: (a) bathyal sediments deposited during Chrons C26n-C24.n.2n in DSDP site 550; (b) shallow marine-fluviatile sediments deposited during Chron C24r in several boreholes from the London area: (c) mid shelf Chrons C26r, C26n, C25r and C24r sediments in the Hales borehole, Norfolk. The higher/lower inclination values correspond relate to Torsvik *et al.* (2001)/Riisager *et al.* (2002) apparent poles.

Clay Formation (middle shelf facies), Woolwich and Reading Beds (fluvial-deltaic/marginal marine facies), and Upnor Formation (shallow marine facies) drilled in the London area of southeast England. This suite of sediments was deposited over a period of about one million years during Chrons C25n and early C24r. Fig. 8(c) shows the inclination data for samples with stable-endpoint directions. As with DSDP Hole 550, the specimens show inclinations clearly lower than the predicted absolute values (58.5°/53.7°).

CAUSE OF THE SHALLOW INCLINATIONS RECORDED AT SHEPPEY

The pattern of anomalously shallow inclinations recorded in lower Palaeogene sedimentary sequences, from a variety of depositional environments, in NW Europe indicate that the phenomenon at Sheppey is not a local or facies-specific feature. Anomalously low inclinations have been encountered in numerous studies of sedimentary rocks, and a number of explanations have been proposed to account for the behaviour including (i) tectonic motion between the sites of the reference pole and the studied sediments (e.g. Cogne *et al.* 1999); (ii) the geomagnetic field has not always behaved as a simple dipole (e.g. Si & Van der Voo 2001); (iii) the depositional process and/or compaction of the sediment pile that has flattened the inclinations (e.g. Rees 1961; Stamathos *et al.* 1989; Arason & Levi 1990; Deamer & Kodama 1990; Dupont-Nivet *et al.* 2002).

In evaluating the Sheppey data, the first two explanations can be excluded. First Sheppey and the NAIP are in tectonic continuity and there has been no discernible convergence between the two sites since the Eocene (actually since the Caledonian orogeny). Secondly, if an anomalous field occupied NW Europe in the early Palaeogene, it would have had to been reasonably long lived and to have had a spectacular configuration in order to create the inclination differences in rocks roughly only 8–10° along a great circle from one another. Deposition-current induced shallowing can also be excluded (Rees 1961). The rocks at Sheppey comprise fine-grained sediments that were thoroughly bioturbated (King 1984, p. 145), and this activity would have completely disturbed a current deflected magnetization. Thus sediment compaction is the favoured explanation. Based on the work of Middleton & Nelson (1996), nodule cementation could have occurred up to a few 100 ka after deposition, thus the sediment that was to form a cemented band had an appreciable time for squash loading to have taken place before the remanence was set.

CONCLUSIONS

In an attempt to increase the database of apparent palaeomagnetic poles used to position Eurasia in the early Cenozoic an apparently high-quality palaeomagnetic pole (Q-factor = 5) has been generated from mid-late Ypresian cemented fine-grained mid-shelf sediments from southeast England. The mean declination of the London Clay Formation sites on Sheppey suggests negligible rotation of NW Europe since the early Eocene. The direction contrasts with a recently derived pole from a systematically studied series of 58–55 Ma NAIP basaltic lavas in the Faroe Islands (Riisager *et al.* 2002), which suggests a moderate clockwise rotation of the region.

The mean inclination value of 43.2° is appreciably lower than the 58.6°/53.8° values predicted by the NAIP derived poles. Compaction flattening is the likely explanation for the anomalously low values; tectonic and geomagnetic explanations can be dismissed. Finally, the fact that inclination shallowing can be shown to affect sedimentary rocks deposited in a wide variety of environments suggests that plate modelling for earlier periods in Earth's history needs to be done carefully when the data is based exclusively on studies of sedimentary rocks.

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